

Tangent-Cocycle Diagnostics for a Chiral Monitored Domain Wall

Asadullah Bhuiyan
(Dated: August 13, 2026)

This note defines the numerical object used to test whether a topological domain wall in the class-A adaptive fermion circuit carries a coherent chiral channel. The object is the differential of the *normalized, record-resolved* Gaussian state update, including both projective measurement and conditional gain/loss reset. It is not the singular-value spectrum of a cumulative Kraus operator, a Choi-covariance transfer chart, or the trajectory-averaged channel. We give the exact fixed-record differential, its stable QR restriction to two predetermined Hamiltonian edge modes, and observables for survival, isotropy, wall retention, directed propagation, and leakage of the encoded perturbation into the classical record. We then state a preregistered numerical campaign and a staged claim criterion. Passing the tangent-channel tests would motivate a separate coherent-information erasure test; it would not by itself establish a quantum error-correcting code. No numerical result is asserted here.

CONTENTS

I. Question and object dictionary	1
II. Exact fixed-record branch differential	2
A. Measurement and reset	2
B. Chronology and covariance perturbations	2
III. A QR-stabilized physical edge channel	3
A. Predetermined edge frame	3
B. Survival, isotropy, and spatial support	3
C. Directed propagation from interference	4
IV. Does the counting record learn the edge coordinate?	4
V. Scattering interpretation and its limit	5
VI. Numerical protocol and validation	5
VII. Staged interpretation and claim gates	6
VIII. Conclusion	6
References	6

I. QUESTION AND OBJECT DICTIONARY

The adaptive preparation protocol of Ref. [1] is a physical quantum instrument: the state is measured in a spatially structured, overcomplete one-particle frame and a record-dependent gain or loss operation corrects undesired occupations. A domain wall between dynamical phases therefore poses an operational question. Does the complete measurement–correction cycle transmit a two-dimensional space of single-particle perturbations along the wall without learning which perturbation was injected?

Ordinary square transfer matrices are poorly conditioned for this question. Exact projectors and gain/loss operations create null directions; in a Choi particle or hole chart these appear as zero or infinite rapidities. Those endpoints are meaningful data about a singular Gaussian operator, but they can obscure the finite channel that acts on a state perturbation. The differential of the physical covariance update instead follows the sensitivity of the normalized trajectory itself. It is finite on every nonzero-probability branch, although it may be rank deficient. Table I records the distinctions that will be maintained throughout.

TABLE I. Related but inequivalent transfer objects.

Object	Operational question	Present use
Choi/Kraus transfer data	What singular sectors does the realized operator K_ξ possess?	Formal representation of projections and gain/loss; not the numerical definition below.
Tangent cocycle	How does $G_0 + \epsilon \delta G_0$ change along the same record?	Primary two-mode transmission diagnostic.
Mean channel	How does $\sum_\xi K_\xi \rho K_\xi^\dagger$ evolve after discarding the record?	Control only; it does not determine quenched trajectory sensitivity.
Flagged instrument	Can a record-dependent recovery preserve arbitrary superpositions?	Eventual error-correction question, tested only after the tangent gate.

In particular, the quenched product of branch differentials cannot be replaced by the differential of an outcome-averaged map. Nor does a slowly contracting tangent direction automatically define a logical degree of freedom: it may be a classical metastable coordinate that is continuously revealed by the counting record.

II. EXACT FIXED-RECORD BRANCH DIFFERENTIAL

A. Measurement and reset

Write the number-conserving covariance as $G = 2\langle c^\dagger c \rangle - \mathbb{I}$, and let $P = |\chi\rangle\langle\chi|$ and $Q = \mathbb{I} - P$ for a normalized measured mode. The outcome $s = +1$ denotes occupied and $s = -1$ empty. Its conditional Born probability and normalized covariance update are

$$p_s(G) = \frac{1 + sg}{2}, \quad g = \langle\chi|G|\chi\rangle, \quad (1)$$

$$F_s(G) = sP + Q(\mathbb{I} + sGP)^{-1}GQ. \quad (2)$$

For a *fixed* outcome s , differentiation gives

$$DF_s(G)[\delta G] = L_s(G) \delta G L_s(G)^\dagger, \quad L_s(G) = Q(\mathbb{I} + sGP)^{-1}. \quad (3)$$

One may multiply L_s by an overall phase; in particular, the minus sign used by a one-particle implementation has no effect on the covariance congruence. Equation (3), rather than a finite difference between two independently sampled outcomes, is the branch differential. The discrete draw s is held fixed.

A conditional swap with an ancilla resets the same mode to covariance $r = +1$ (filled) or $r = -1$ (empty):

$$R_r(G) = QGQ + rP, \quad DR_r(G)[\delta G] = Q \delta G Q. \quad (4)$$

Thus a corrected event has the column factor

$$L_{s,r}^{\text{event}}(G) = QL_s(G). \quad (5)$$

For an ideal rank-one measurement and a reset of exactly the same normalized mode, $QL_s = L_s$. Omitting a separate Q multiplication in this specific branch therefore gives the same Jacobian in exact arithmetic; legacy GPU spectra are not invalidated for this reason alone. New code and tests nevertheless represent the reset explicitly, both to audit chronology and to avoid assuming this identity for generalized supports or imperfect mode matching.

Equations (2)–(5) are evaluated only when $p_s(G) > 0$. An exactly forbidden forced branch is invalid; replacing its singular denominator by a numerical regulator and calling the result a physical Lyapunov step changes the problem. Production data must therefore save the minimum Born denominator, null ranks, and the number of invalid forced branches.

B. Chronology and covariance perturbations

Let $L_{t,j}$ be the factor for event j of cycle t , including the reset when it occurs, and define the chronological cycle factor

$$\mathcal{L}_t = L_{t,M_t} \cdots L_{t,2} L_{t,1}. \quad (6)$$

For a realized record ξ , the restricted one-particle product and the full covariance tangent are

$$\mathcal{A}_T(\xi) = \mathcal{L}_T \cdots \mathcal{L}_2 \mathcal{L}_1, \quad (7)$$

$$\delta G_T = \mathcal{A}_T \delta G_0 \mathcal{A}_T^\dagger. \quad (8)$$

This convention puts the factor acting on perturbation columns on the left. It is equivalent, after adjoining the one-particle transfer convention, to the usual Gaussian Schur-complement differential. Its Lyapunov exponents are quenched singular-value growth rates of a state derivative, not many-body excitation gaps and not eigenvalues of an averaged Lindbladian.

III. A QR-STABILIZED PHYSICAL EDGE CHANNEL

A. Predetermined edge frame

The input frame is chosen before any monitored trajectory is generated. Let H_{DW} be the exact or spectrally flattened single-particle domain-wall Hamiltonian used to define the protocol. On a chosen wall, take two orthonormal edge-band modes f_0, f_1 at adjacent momenta nearest the edge crossing and set

$$W_0 = (f_0, f_1), \quad W_0^\dagger W_0 = \mathbb{I}_2. \quad (9)$$

The localization and energy-window rules, including tie breaking, are fixed once for every system size. The two modes are not learned from the slowest Lyapunov vectors. This prevents the diagnostic from manufacturing a channel by optimizing its input on the same disorder records used for evaluation.

Directly multiplying $\mathcal{A}_T W_0$ underflows or overflows. Starting from $Q_0 = W_0$ and $B_0 = \mathbb{I}_2$, perform one thin QR factorization per full cycle,

$$Y_t = \mathcal{L}_t Q_{t-1} = Q_t R_t, \quad B_t = R_t B_{t-1}. \quad (10)$$

Then the identity

$$\mathcal{A}_t W_0 = Q_t B_t \quad (11)$$

is exact up to roundoff. Stabilize the 2×2 coordinate matrix by writing $B_t = e^{\Gamma_t} \widehat{B}_t$, adding a scalar log norm to Γ_t after each multiplication. Singular values are reconstructed as $\log \sigma_i(B_t) = \Gamma_t + \log \sigma_i(\widehat{B}_t)$. An exact zero singular value is recorded as a null channel, not silently clipped into a finite exponent.

B. Survival, isotropy, and spatial support

Let $\sigma_1 \geq \sigma_2 \geq 0$ be the singular values of B_T . The common survival exponent and the scale-free anisotropy are

$$\bar{\lambda}_T = \frac{\log \sigma_1 + \log \sigma_2}{2T}, \quad (12)$$

$$\delta_{\text{iso}}(T) = \frac{\sigma_1^2 - \sigma_2^2}{\sigma_1^2 + \sigma_2^2} = \tanh \left| \log \frac{\sigma_1}{\sigma_2} \right|. \quad (13)$$

A neutral but strongly anisotropic channel preserves at most one direction; it cannot carry an arbitrary two-component perturbation. Both quantities are invariant under a $U(2)$ change of the initial edge basis.

For a real-space projector Π_w onto a fixed-width strip around the chosen wall, define

$$\eta_w(t) = \frac{\|\Pi_w Q_t B_t\|_F^2}{\|Q_t B_t\|_F^2}, \quad \eta_{w \rightarrow \bar{w}}(t) = \frac{\|\Pi_{\bar{w}} Q_t B_t\|_F^2}{\|Q_t B_t\|_F^2}. \quad (14)$$

The common scalar e^{Γ_t} cancels. The strip width is specified in lattice units before the run and held fixed in the size extrapolation. The first ratio measures wall retention and the second leakage to the opposite wall; their sum need not equal one because there is also intervening bulk.

C. Directed propagation from interference

Contraction alone does not establish chirality. For adjacent momenta, inject $u_+ = (1, e^{i\phi_0})^T/\sqrt{2}$ and form the normalized propagated density

$$\rho_t(x, y, \mu) = \frac{|[Q_t \hat{B}_t u_+]_{x,y,\mu}|^2}{\|Q_t \hat{B}_t u_+\|^2}. \quad (15)$$

Its first circular moment along the periodic direction is

$$Z_t = \sum_{x,y,\mu} e^{2\pi i y/N_y} \rho_t(x, y, \mu), \quad y_c(t) = \frac{N_y}{2\pi} \text{unwrap arg } Z_t. \quad (16)$$

The slope of $y_c(t)$ after burn-in defines a trajectory velocity. The analysis is repeated with several fixed ϕ_0 values to exclude a phase-choice artifact. A chiral domain-wall interpretation requires nonzero mean velocities with opposite signs on the two oppositely oriented walls, while randomized and raster schedules quantify ordering bias.

IV. DOES THE COUNTING RECORD LEARN THE EDGE COORDINATE?

Near a maximally mixed reference, a two-mode Hermitian perturbation can be written

$$\delta G_a(0) = W_0 \sigma_a W_0^\dagger, \quad a \in \{x, y, z\}, \quad (17)$$

where σ_a are Pauli matrices. Propagate each generator by the same fixed record using Eq. (8). Immediately before measurement event j , set

$$d_{j,a} = \langle \chi_j | \delta G_a(j^-) | \chi_j \rangle, \quad g_j = \langle \chi_j | G(j^-) | \chi_j \rangle. \quad (18)$$

The realized score and the conditional classical Fisher increment are then

$$h_{j,a} = \partial_a \log p_{s_j} = \frac{s_j d_{j,a}}{1 + s_j g_j}, \quad (19)$$

$$J_{ab}^{(j)} = \sum_{s=\pm 1} p_s h_{j,a}(s) h_{j,b}(s) = \frac{d_{j,a} d_{j,b}}{1 - g_j^2}, \quad (20)$$

for $|g_j| < 1$. Deterministic endpoints are evaluated by the corresponding physical limit rather than by dividing by zero. Because conditional score increments form a martingale difference, the expected Fisher information of the complete record is the expected sum of Eq. (20). The principal statistic is its largest eigenvalue per observed cycle.

This quantity supplies the missing “record privacy” test. A small δ_{iso} says that the normalized tangent map treats two input directions similarly; it does not say that their Born weights are the same. A neutral wall coordinate with finite Fisher-information density is therefore a long-lived *classical measured mode*, not a coherent memory.

For comparison, append the classical flag to a fixed-record instrument,

$$\mathcal{I}(\rho) = \sum_{\xi} |\xi\rangle\langle\xi| \otimes K_{\xi} \rho K_{\xi}^\dagger. \quad (21)$$

If recovery may depend on the accessible flag, exact Knill–Laflamme correctability of a code projector P_V requires, branch by branch,

$$P_V K_{\xi}^\dagger K_{\xi} P_V = p_{\xi} P_V. \quad (22)$$

The record Fisher test probes the infinitesimal state-independence of p_{ξ} , while two-mode isotropy probes part of the normalized branch action. Neither test alone, nor their conjunction at finite size, proves Eq. (22) or constructs a recovery map [2].

V. SCATTERING INTERPRETATION AND ITS LIMIT

Separate common amplification from channel anisotropy by defining

$$g_T = (\sigma_1 \sigma_2)^{1/2}, \quad \tilde{\sigma}_{1,2} = \sigma_{1,2}/g_T = e^{\pm x_T}, \quad x_T = \frac{1}{2} \log \frac{\sigma_1}{\sigma_2}. \quad (23)$$

Then $\delta_{\text{iso}} = \tanh(2x_T)$. In a reciprocal flux-conserving two-port transfer problem, the usual polar decomposition identifies $\tau = \text{sech}^2 x_T$ as a transmission eigenvalue [3, 4]. This supplies a useful dictionary: the QR cocycle measures normalized sensitivity transmitted from one temporal boundary to the other, while x_T is its imbalance rapidity.

The equality with a physical particle-transmission probability is *conditional*. A monitored gain/loss tangent need not preserve a flux metric or admit the reciprocal singular-value pairing of a unitary scattering network. Without an explicit current-conserving dilation, $\text{sech}^2 x_T$ is a transmission *proxy*, not an S -matrix probability. Conversely, the divergent directions of a singular Choi transfer chart should not be inserted into Eq. (23): they encode exact occupied/empty caps of the Kraus relation, whereas the present σ_i describe the differential of a normalized state trajectory.

VI. NUMERICAL PROTOCOL AND VALIDATION

The primary data are Born-sampled, perfect-correction trajectories of the canonical adaptive circuit. Postselection is retained only as a deterministic implementation check. The planned campaign is summarized in Table II; these are prospective parameters, not completed results. This note specifies intended behavior; implementation and completed-run status must be established separately by source, tests, and output metadata.

TABLE II. Prospective steady-ensemble campaign.

Quantity	Choice
Geometry	$N_x = 20, N_y \in \{24, 32, 48, 64, 80\}$
Frame truncation	$n_{\text{shell}} = 1$; repeat $N_y = 32, 64$ at $n_{\text{shell}} = 2$
Time window	burn-in $2N_y$, observation $4N_y$ cycles
Domain-wall schedule	symmetric randomized ordering
Couplings	$\alpha_{\text{in}} = 1, \alpha_{\text{out}} = 30$, wall truncation on
Sampling	128 trajectories for primary and uniform control; 64 otherwise
Primary feedback	Born-sampled perfect correction
Validation branch	forced postselection, never substituted for the primary ensemble
Controls	uniform topological, trivial hard wall with the same mask, same-Chern artificial cut, and raster ordering

Before data from this protocol are used for the staged claims of Sec. VII, the corresponding implementation must pass the following tests.

1. For a forced nonzero-probability branch, compare

$$\frac{F_{\xi}(G + \epsilon \delta G) - F_{\xi}(G - \epsilon \delta G)}{2\epsilon} \quad \text{with} \quad \mathcal{A}_{\xi} \delta G \mathcal{A}_{\xi}^{\dagger} \quad (24)$$

over a convergence window in ϵ . Cover mixed and nearly pure G , occupied and empty outcomes, mismatch and no-mismatch branches, gain and loss, and local and dense update paths.

2. Represent Eq. (4) explicitly in the CPU/GPU audit trail and test $QL_s = L_s$ for the current branch. Under identical forced records, compare final covariance, event weights, QR factors, null counts, and restricted singular values.
3. Check Eq. (11) against an unstabilized small-system product, and verify independence under QR frequency and an arbitrary initial $U(2)$ rotation of W_0 .
4. Save denominator minima and censor invalid zero-probability branches rather than regularizing them into the physical sample.

VII. STAGED INTERPRETATION AND CLAIM GATES

The finite-size analysis uses trajectory bootstrap confidence intervals and fits versus N_y that retain raw null/censor counts. The staged gate is:

1. The magnitudes $|\bar{\lambda}_T|$, δ_{iso} , and the largest record-Fisher eigenvalue per cycle must each decrease with N_y with a positive decay exponent at 95% confidence.
2. At $N_y = 80$, require $\delta_{\text{iso}} < 0.1$ and a lower 95% confidence bound on wall retention above 0.8.
3. On the two walls, the velocity confidence intervals must exclude zero and have opposite signs.
4. No control may reproduce this entire conjunction.

The numerical thresholds are conservative preregistration choices, not universal constants. Failure has a useful interpretation: depending on which condition fails, the wall supports a localized slow tangent sector, a directed but classically monitored perturbation, or a one-dimensional survivor, but not the proposed coherent two-mode channel.

Only if all four conditions pass do we proceed to a Song-type erasure test for a fixed-charge Gaussian candidate code

$$\mathcal{V} = \text{span} \left\{ f_0^\dagger \prod_{j=1}^{N-1} d_j^\dagger |0\rangle, f_1^\dagger \prod_{j=1}^{N-1} d_j^\dagger |0\rangle \right\}. \quad (25)$$

Arbitrary superpositions in this two-dimensional space remain Slater determinants because only one occupied orbital is varied. For $\bar{\rho} = P_V/2$ and a local erased region A , the coherent- information loss is

$$\Delta(A; \mathcal{V}) = S_A(\bar{\rho}) + \log 2 - S_{\bar{A}}(\bar{\rho}). \quad (26)$$

Its decay with distance and erased length gives a direct approximate-recovery diagnostic in the spirit of chiral-edge and CFT codes [5, 6]. The mixture $\bar{\rho}$ is generally non-Gaussian even though both codewords are Gaussian; its entropy cannot be computed from the averaged covariance alone. This gated calculation must be validated against exact small-system Fock-space reductions.

This proposal is adjacent to dynamically generated and Floquet codes, where a measurement schedule transports a genuine logical algebra and may enforce anomalous boundary flow [7, 8]. The present adaptive circuit instead begins as dissipative state preparation. Until Eq. (22), coherent-information recovery, or an equivalent operator-algebra test is demonstrated, the justified language is *chiral tangent channel* or *boundary anomaly of active stabilization*, not *dynamical edge code*.

VIII. CONCLUSION

The proposed calculation replaces a singular transfer-core diagnostic by the well-defined differential of the actual measurement-feedback trajectory. Its QR restriction asks a deliberately narrow question: do two physically chosen Hamiltonian edge modes survive, remain isotropic and wall-localized, move chirally, and remain hidden from the Born record? The combined answer is more discriminating than a near-zero Lyapunov exponent. It also creates a clean decision boundary between three possible outcomes—a slow classical wall coordinate, a coherent chiral tangent channel, and a genuine approximate edge code—without presupposing that the strongest interpretation is true.

-
- [1] A. H. Bhuiyan, H. Pan, and C.-M. Jian, *Physical Review Research* **8**, 023147 (2026), [arXiv:2507.13437 \[cond-mat.str-el\]](#).
 - [2] E. Knill and R. Laflamme, *Physical Review A* **55**, 900 (1997), [arXiv:quant-ph/9604034](#).
 - [3] J. T. Chalker and P. D. Coddington, *Journal of Physics C: Solid State Physics* **21**, 2665 (1988).
 - [4] C. W. J. Beenakker, *Reviews of Modern Physics* **69**, 731 (1997), [arXiv:cond-mat/9612179](#).
 - [5] Y. Song, Z. Liu, Z. Wang, J. Y. Lee, and B. Shi, Approximate quantum error correction at chiral topological edges (2026), [arXiv:2608.06258 \[quant-ph\]](#).
 - [6] S. Sang, T. H. Hsieh, and Y. Zou, Approximate quantum error correcting codes from conformal field theory (2024), [arXiv:2406.09555 \[quant-ph\]](#).
 - [7] M. B. Hastings and J. Haah, *Quantum* **5**, 564 (2021), [arXiv:2107.02194 \[quant-ph\]](#).
 - [8] D. Aasen, J. Haah, Z. Li, and R. S. K. Mong, Measurement quantum cellular automata and anomalies in floquet codes (2023), [arXiv:2304.01277 \[quant-ph\]](#).